Cascaded Speed and Torque Control of a 6/4 Switched Reluctance Motor with Cubic Torque Sharing and Independent Python Reproduction

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***Abstract*—This paper consolidates the final deliverable of an electric-drives project on a three-phase 6/4 switched reluctance motor (SRM) into a traceable and reproducible form. The implemented drive combines an outer proportional–integral speed regulator, a cubic torque sharing function (TSF), nonlinear torque-to-current inversion, hysteresis current control, and an asymmetric half-bridge converter inside a MATLAB/Simulink plant model. The key physical design choice is the cubic TSF, whose zero-slope commutation boundaries remove the reference steps created by simpler linear sharing laws and therefore reduce current and torque discontinuities at phase handover. Five operating cases cover nominal tracking, doubled-speed operation, a torque-limit load step, a speed ramp, and a load ramp. The rerunnable repository results show exact or near-exact steady-state tracking in the nominal, high-speed, and ramp cases, while the maximum-load case exposes the expected saturation boundary rather than an instability. An independent Python re-implementation confirms reproducibility against the archived Simulink validation metrics, with differences limited to the quantities most sensitive to solver order and time discretization. The result is a complete, defensible baseline for future SRM work rather than a collection of waveform screenshots.**

***Index Terms*—switched reluctance motor, cubic torque sharing, hysteresis current control, reproducibility**

I. Introduction

Switched reluctance motors (SRMs) remain attractive wherever ruggedness, thermal tolerance, and freedom from rare-earth magnets matter, but they trade mechanical simplicity for control difficulty. Torque is generated by a strongly position-dependent inductance rather than by sinusoidal back-EMF, so commutation, current control, and speed control must all cooperate tightly with rotor position [1]–[3].

The literature already establishes the importance of current profiling and torque sharing functions for ripple reduction [4]–[7]. What is less common in course-scale or open project repositories is a single deliverable that connects four pieces at once: the full controller stack, the physical reason for the TSF choice, a compact but meaningful operating-envelope study, and an independent rerun that checks whether the archived results are reproducible rather than merely plausible.

This paper addresses that gap for a three-phase 6/4 SRM project completed in MATLAB/Simulink and mirrored in Python. The work is organized around four claims that are directly supported by the repository artifacts. First, the project implements the full cascaded PI–TSF–T2I–hysteresis drive with an asymmetric half-bridge converter for a 6/4 machine. Second, the cubic TSF matters physically because its zero boundary slopes remove commutation-edge reference steps that a linear sharing law would inject. Third, the drive performs robustly across the practical operating envelope, while the most severe load case reveals the saturation limit in a physically explainable way rather than through unstable behavior. Fourth, an independent Python mirror reproduces the archived Simulink behavior closely enough to support trust in the model, with the remaining gaps tied to solver order and time discretization.

II. Drive Formulation

A. Machine Model

The machine is a three-phase 6/4 SRM with six stator poles, four rotor poles, and concentrated stator windings grouped into the phase pairs shown in Fig. 1. The adopted parameters are $R_s = 1.0\ \Omega$, $L_{\min} = 20$ mH, $L_{\max} = 150$ mH, $J = 0.002\ \mathrm{kg\,m^2}$, $B = 0.001\ \mathrm{N\,m\,s/rad}$, and $V_{dc} = 300$ V.

Under the standard negligible-mutual-coupling assumption, each phase obeys

$$v_k = R_s i_k + L(\theta)\frac{di_k}{dt} + i_k \omega \frac{dL}{d\theta}, \tag{1}$$

with electromagnetic torque

$$T_{e,k} = \frac{1}{2} i_k^2 \frac{dL_k}{d\theta}, \tag{2}$$

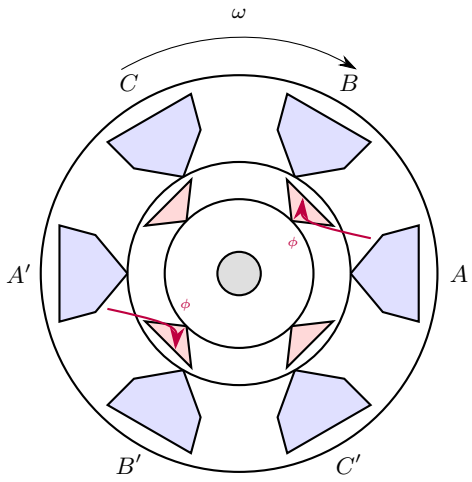


Fig. 1. 6/4 SRM cross-section with three stator phases and four rotor poles. Opposite stator teeth form one phase pair, while torque is produced as the rotor moves toward the energized low-reluctance path.

and mechanical dynamics

$$J\frac{d\omega}{dt} = \sum_{k=A,B,C} T_{e,k} - T_L - B\omega. \tag{3}$$

Positive motoring torque therefore exists only while $dL/d\theta > 0$. That simple sign condition is what makes commutation timing and current decay decisive in SRM drives [2], [3].

B. Controller Stack

The implemented controller follows the cascade in Fig. 2. The outer loop produces total torque demand

$$T^*(s) = K_p e_\omega(s) + \frac{K_i}{s} e_\omega(s), \tag{4}$$

with $K_p = 0.5$, $K_i = 10\ \mathrm{s}^{-1}$, and a torque clamp of $T_{\max} = 5\ \mathrm{N\,m}$.

That total demand is distributed among active phases and inverted through

$$i_k^* = \sqrt{\frac{2T_k^*}{dL_k/d\theta}}, \tag{5}$$

whenever $dL_k/d\theta > 0$, with current limited to $I_{\max} = 15$ A. The inner hysteresis law then applies $+V_{dc}$, 0, or $-V_{dc}$ across each phase according to the current error band $\Delta i = 0.1$ A. This is the practical role of the asymmetric half-bridge: fast magnetization, freewheeling around zero voltage, and active demagnetization before the phase enters the negative-inductance region [5].

C. Why the Cubic TSF Was Chosen

The adopted sharing law is

$$f(x) = 3x^2 - 2x^3, \qquad x = \frac{\theta - \theta_{\mathrm{on}}}{\theta_{\mathrm{ov}}} \in [0,1]. \tag{6}$$

Its practical value is not that it is the only smooth TSF available, but that it removes the two most damaging reference discontinuities with minimal implementation cost. Because $f'(0) = f'(1) = 0$, the incoming phase current starts from zero slope and the outgoing phase current decays to zero slope. Compared with a linear sharing law, that means the hysteresis controller is not asked to chase an instantaneous current-reference jump at either commutation edge. The smoother current handover is exactly the mechanism behind reduced torque discontinuity.

TABLE I
KEY PARAMETERS AND TEST MATRIX

Quantity	Value
R_s	1.0 Ω
$L_{\min}, L_{\max}$	20 mH, 150 mH
J, B	0.002 kg m^2, 0.001 N m s/rad
V_{dc}, $I_{\max}$	300 V, 15 A
$T_{\max}$, Δi	5 N m, 0.1 A
TC1	100 rad/s, 1.5 N m step at 0.1 s
TC2	200 rad/s, 1.5 N m
TC3	100 rad/s, 5 N m step at 0.1 s
TC4	0 to 100 rad/s ramp
TC5	100 rad/s with 0 to 5 N m load ramp

III. Simulation Setup and Evidence Sources

The project repository contains two evidence layers. The first is the archived course implementation in MATLAB/Simulink [8]: the Simulink model (`src/SRM_Project.slx`), the original project report, and the back-to-back validation report shipped under `docs/`. The second is the rerunnable Python mirror (`validation/srm_validation.py`), which reproduces the same controller logic with a fixed-step forward-Euler solver. To keep the paper reproducible directly from the repository, the plotted result figures in this manuscript are regenerated from the Python mirror using the shared parameter set, while the Simulink side enters quantitatively through the archived validation metrics.

The common simulation settings are a fixed time step of $T_s = 10\ \mu$s, a total window of 0.5 s, turn-on angle $\theta_{\mathrm{on}} = 0°$, turn-off angle $\theta_{\mathrm{off}} = 38°$, and overlap angle $\theta_{\mathrm{ov}} = 10°$. The nominal outer-loop gains are $K_p = 0.5$ and $K_i = 10\ \mathrm{s}^{-1}$, with the ramp-tracking case retuned to $K_p = 15$ and $K_i = 10\ \mathrm{s}^{-1}$.

IV. Results

A. Representative Baseline Case

Figure 4 is the reference case against which the rest of the envelope should be read. The rerun reaches 100 rad/s in 32.4 ms, settles to 99.998 rad/s over the last fifth of the window, and produces 1.600 N m mean electromagnetic torque in steady state. Those numbers matter because they separate the two jobs of the controller. During startup, the torque clamp forces the drive to accelerate with maximum available authority. After the 1.5 N m load step at 0.1 s, the outer PI loop only has to replenish the demanded load torque and viscous loss, so the current envelopes collapse from the startup transient to the much smaller steady-state pulses visible in the middle trace. The smooth phase handover is the practical manifestation of the cubic TSF argument from Fig. 3.

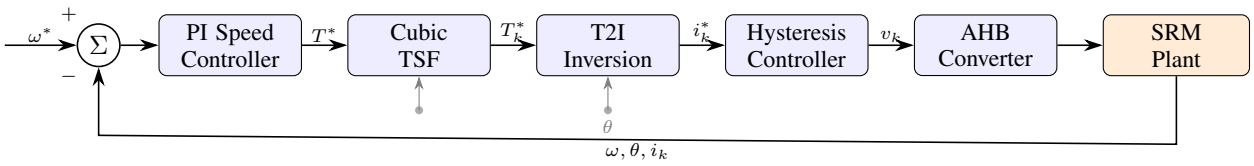


Fig. 2. Full cascaded drive architecture implemented in the project. The outer PI loop generates total torque demand, the cubic TSF allocates phase torques, the T2I block maps them to current references, the hysteresis loop switches the asymmetric half-bridge, and the SRM plant feeds back speed, angle, and currents.

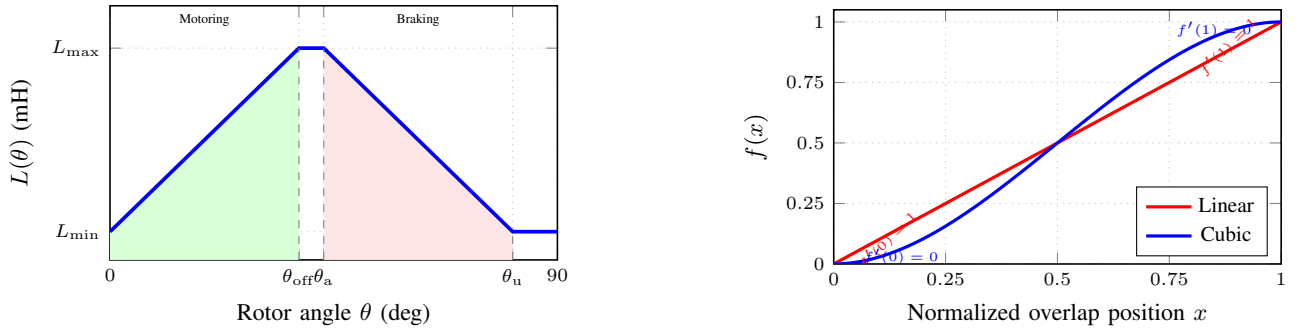


Fig. 3. Theory figure with direct physical payoff. Left: the 6/4 SRM produces positive torque only in the rising-inductance interval. Right: the selected cubic torque sharing law $f(x) = 3x^2 - 2x^3$ has zero slope at both overlap boundaries, unlike the linear law, so the current reference enters and exits commutation smoothly instead of with a step.

B. Operating Envelope

Figure 5 replaces four nearly identical waveform dumps with four different questions. Panel (a) asks whether the same controller still works when motional back-EMF doubles. The answer is yes: TC2 settles to 199.999 rad/s and its steady-state torque remains positive throughout the last fifth of the run, varying between 1.37 and 2.11 N m instead of crossing into braking torque. That behavior occurs because the AHB and hysteresis loop still demagnetize each phase before the inductance slope turns negative, even though the electrical window is tighter at higher speed.

Panel (b) asks what happens when the load step reaches the controller limit. TC3 does not hide that boundary. After the 5 N m step, the mean electromagnetic torque rises to 5.01 N m but the speed settles near 79.6 rad/s rather than returning to 100 rad/s. That is not a control failure. It is the expected consequence of testing exactly at the torque clamp, because the controller must cover the commanded load and viscous loss with the same 5 N m ceiling. The important result is graceful degradation: the drive stays stable, the commutation remains well ordered, and the model makes the limit visible instead of masking it.

Panel (c) asks whether the cascade can follow a dynamic speed command without becoming noisy. With K_p increased to 15, TC4 tracks the 0 to 100 rad/s ramp with 0.214 rad/s root-mean-square error and 1.25 rad/s maximum absolute error over the ramp interval. The improvement is causal. The ramp case needs more proportional authority than the step-tuned nominal controller, because a type-1 loop can remove constant error exactly but leaves finite ramp error unless the outer loop gain is increased.

Panel (d) asks whether the drive can increase torque continuously instead of reacting only to steps. In TC5, the electromagnetic torque follows the ramped load while the mean speed over the final fifth of the run remains 98.995 rad/s. At the electrical level, the peak phase current grows from about 4.5 A soon after the ramp starts to about 7.2 A near the end. That monotonic current growth is the expected signature of the nonlinear T2I block doing its job across the torque range, not just around one operating point.

V. Independent Reproducibility Check

Figure 6 addresses the question that matters most for trust in a custom course project: are the results reproducible outside the original Simulink canvas? The answer is yes for the quantities that should survive an implementation change. The archived Simulink TC1 reference gives 99.94 rad/s steady-state speed, 29.4 ms rise time, 1.601 N m mean steady-state torque, and 7.21 A peak phase current. The current Python rerun gives 99.998 rad/s, 32.4 ms, 1.600 N m, and 7.269 A, respectively. In relative terms that means 0.06% speed difference, 0.09% mean-torque difference, and 0.82% peak-current difference.

The only visible transient gap is rise time, where the rerun is 3.0 ms slower. That difference is physically explainable. The Simulink reference was generated with fixed-step fourth-order Runge–Kutta, whereas the repository mirror integrates with forward Euler. Both solve the same electromechanical model, but the lower-order solver resolves the fast startup current build-up less aggressively. The archived validation report also flagged torque-ripple detail as the most solver-sensitive quantity, which is consistent with the fact that ripple depends strongly on how the inductance gradient is discretized

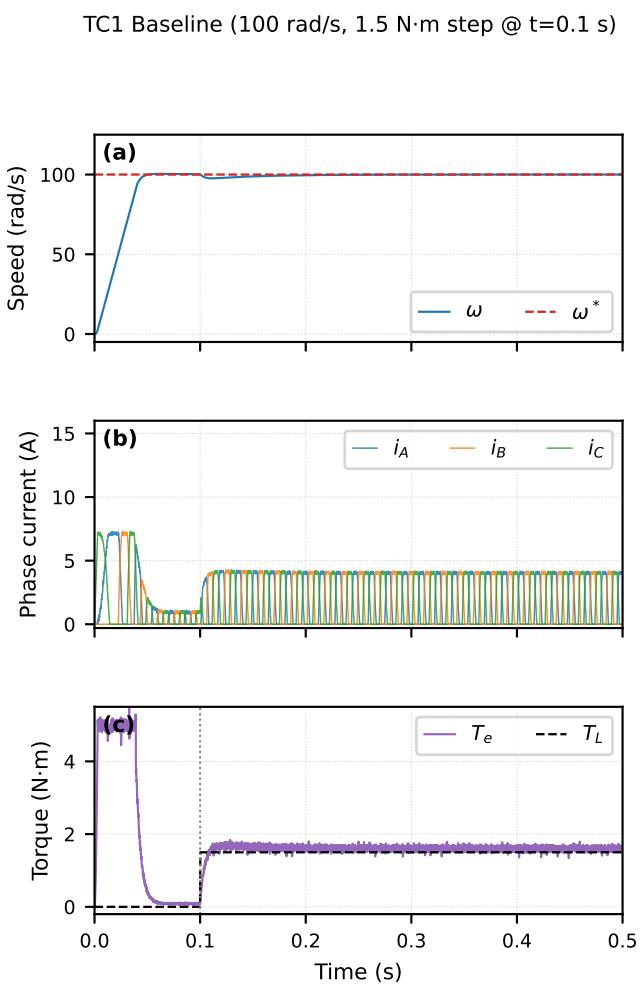


Fig. 4. Representative baseline response for TC1. The speed loop reaches the 100 rad/s command, the cubic TSF shapes the three phase currents through commutation, and the electromagnetic torque settles to the 1.5 N m load after startup saturation clears.

and on how accurately the switching instants are resolved. The important point is not that every transient sample is identical. It is that the independent implementation reproduces the same controller logic, the same steady-state operating point, and the same qualitative dynamics without shared simulation code.

VI. Discussion and Limitations

The evidence set supports a stronger conclusion than a checklist-style report, but it also supports a more honest one. The drive is clearly effective across normal operating conditions, yet the figures also show where the model and controller stop being ideal.

The first important interpretation concerns the cubic TSF. Its benefit is not abstract smoothness for its own sake. It reduces the excitation applied to the hysteresis controller at commutation. Because the current reference arrives with zero edge slope, the current loop responds with overlap pulses that are shaped rather than kicked. That is why the phase-current transitions in Fig. 4 look physically continuous instead of piecewise abrupt.

The second interpretation concerns the envelope itself. TC2 and TC4 show what robust regulation looks like when the plant stays inside the controller's authority. TC3 shows what honest reporting looks like when it does not. The project is stronger for showing the saturation boundary explicitly, because that boundary is exactly what motivates future angle scheduling, higher torque headroom, or softer startup logic.

Three limitations should therefore be kept attached to the project. First, the inductance model is piecewise linear and current independent, so magnetic saturation and deep nonlinear co-energy effects are not represented. Second, the firing angles are fixed, which is acceptable around the studied operating points but suboptimal once speed and back-EMF increase further. Third, the repository preserves the Simulink model and the archived reports but not exported Simulink time-series arrays, so the paper's plotted waveforms are regenerated from the Python mirror and the Simulink side enters numerically through the archived validation metrics. None of these limitations invalidate the current conclusions, but each one defines a clear next refinement step.

VII. Conclusion

This work presents a complete and traceable SRM-drive deliverable built around a three-phase 6/4 machine, a cascaded PI–TSF–T2I–hysteresis controller, and an asymmetric half-bridge converter. The final paper deliberately focuses on evidence rather than volume. The figure set distinguishes machine topology, controller structure, TSF rationale, representative nominal behavior, envelope behavior, and reproducibility.

The resulting picture is coherent. The cubic TSF is physically justified because its zero-slope boundaries reduce commutation discontinuity. The drive tracks nominal, high-speed, and ramped commands tightly, while the 5 N m step case reveals the expected torque-limit boundary without instability. The independent Python mirror reproduces the archived Simulink metrics closely enough to support trust in the model, with the residual differences explained by solver order and numerical modeling choices rather than by contradictory physics.

That combination makes the repository suitable as a final course artifact and as a defensible starting point for future SRM extensions such as firing-angle scheduling, saturation-aware inductance maps, or hardware implementation.

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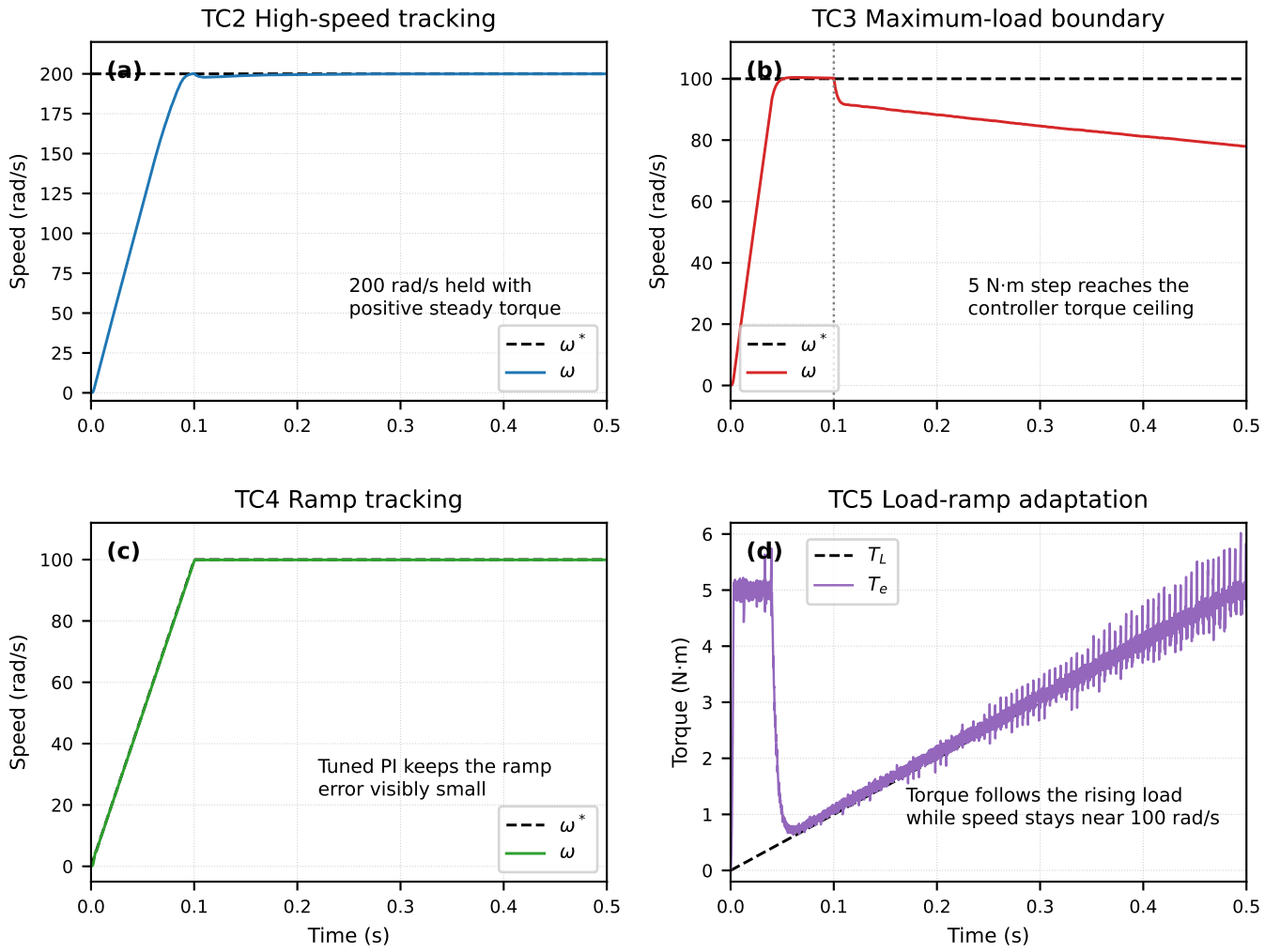


Fig. 5. Operating-envelope composite derived from TC2–TC5. (a) TC2 confirms high-speed tracking at 200 rad/s. (b) TC3 shows the saturation boundary under the maximum 5 N m step load. (c) TC4 demonstrates low-error ramp tracking after retuning the outer loop. (d) TC5 shows torque following a continuous load ramp while speed stays close to the 100 rad/s target.

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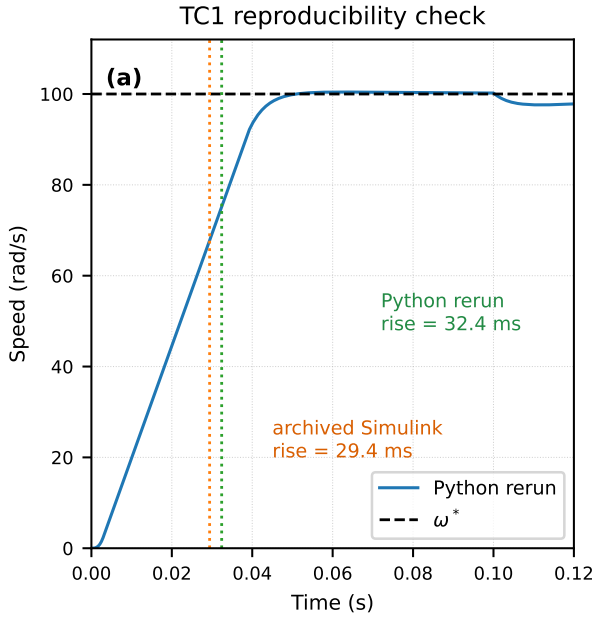
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(b) Archived Simulink vs. Python rerun

Metric	Simulink ref	Python rerun	Δ
Speed (rad/s)	99.94	99.998	0.06%
Rise time (ms)	29.4	32.4	3.0 ms
Mean torque (N·m)	1.601	1.600	0.09%
Peak current (A)	7.21	7.269	0.82%

Source traceability: Simulink values from docs/Validation_Report.pdf; Python values from the current validation/srm_validation.py rerun.

Fig. 6. Independent TC1 reproducibility check. The left panel shows the rerun Python speed trace with the archived Simulink and current Python rise-time markers. The right panel compares archived Simulink metrics from `docs/Validation_Report.pdf` against the current Python rerun from `validation/srm_validation.py`.